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ASSESSMENT-I C01-AT1

1. Breast Cancer classification price:

TO Find False Negatives (FN)

each class = 1800 Samples

Testing Data = 20%.

Test Sample per class = $1800 \times 20\% = 360$

For the positive class:

$$TP + FN = 360$$

$$210 + FN = 360$$

$$FN = 150$$

Given:

$$TP = 210$$

$$TN = 190$$

$$FP = 30$$

$$FN = 150$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{210 + 190}{210 + 190 + 30 + 150} = \frac{400}{580}$$

$$= 0.6897$$

$$= 68.97\%$$

precision:

$$\frac{TP}{TP + FP} = \frac{210}{210 + 30} = \frac{210}{240} = 0.875$$

$$= 87.5\%$$

Sensitivity (Recall)

$$\frac{TP}{TP + FN} = \frac{210}{210 + 150} = \frac{210}{360} = 0.5833$$

$$= 58.33\%$$

Specificity

$$\frac{TN}{TN + FP} = \frac{190}{190 + 30} = \frac{190}{220} = 0.8636$$

$$= 86.36\%$$

Effectiveness in medical diagnosis:

Accuracy : 68.97%

Precision : 87.5%

Sensitivity : 58.33%

Specificity : 86.36%

Conclusion:

The model has high precision and specificity but low sensitivity, making it less suitable for medical diagnosis because missing cancer cases (high FN) can be dangerous.

2. Training and validation loss:

c.i) Identify overfitting:

Epoch	Training Loss	Validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

The model begins to overfit at 4.

c.ii) Reduce overfitting:

* Early Stopping monitors the validation loss during training, when VL starts increasing while the training loss continues to decrease, training is stopped. This prevents the model from learning unnecessary details and noise from the training data.

* Dropout randomly deactivates a certain percentage of neurons during each training iteration. This prevents the model from depending too much on specific neurons, reduces overfitting and helps it learn more general features.

(iii) overfitting affects model's generalization:

It occurs when a model learns the training data too closely, including its noise or unnecessary details, instead of learning general patterns. As a result, the model achieves low training loss and high training accuracy, but performs poorly on new or unseen data because it cannot generalize effectively. This leads to higher validation loss, lower prediction accuracy and reduced reliability in real world applications.